UNIT IV

Chapter 10

Neural Control And Coordination

Chapter Outline

10.1 Neural system

10.2 Human Neural System

10.3 Neuron as a structural and functional unit of neural system

10.4 Central neural system

10.5 Reflex action and reflex arc

10.6 Sensory reception and processing



Gamma-aminobutyric acid, or GABA, is the brain's major inhibitory neurotransmitter that reduces neuronal excitability.

Learning Objectives:

- *Understands the structure of neuron and neural system of human beings*
- *Learns to differentiate the functions of sensory and motor neuron*
- *Understands the conduction of nerve impulses and learns the importance of myelin sheath-saltatory conduction.*
- *Outlines the role of synapse and neuromuscular junction.*
- *Learns the structure and functions of central neural system*
- *Understands the structure, sensory reception and processing in Photo, Phono, Olfactory, Gustatory and Skin Receptors*

Did you ever wonder how our body functions? The body maintains a stable condition even when the outside environment changes; Our eyes help to see things around us; Ears help us to hear various sounds; Heart beats continuously and rhythmically; Air goes in and out of lungs; Eyes shed tears when our limbs get hurt. Each cell of the body works in a coordinated manner. Do you know how it is coordinated and controlled?

The neural system of our body coordinates all the other systems to work together effectively and smoothly. Every second, diverse functions in our body are performed by the neural system. Day and night, millions of messages pass as stimuli through the cells of the neural system to stimulate the heart to beat; kidney to excrete waste; and mouth to relish the delicious food. An even more remarkable

XI Std Zoology Chapter-10 EM.indd 176 30/12/2022 13:55:30